

30 - Hydrocarbons

Reactions of benzene & methylbenzene

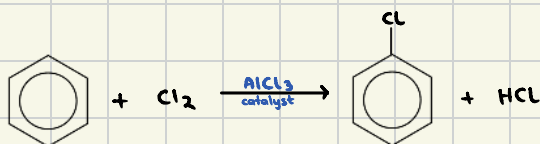
① substitution reactions

→ chlorination of benzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.

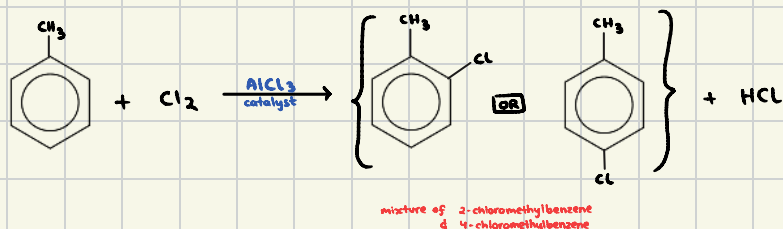


→ chlorination of methylbenzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.

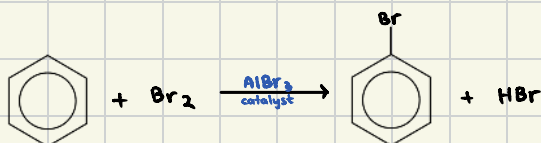


→ bromination of benzene

reagent : bromine (Br_2)

catalyst : AlBr_3 (or Fe)

conditions : anhydrous conditions & room temp.

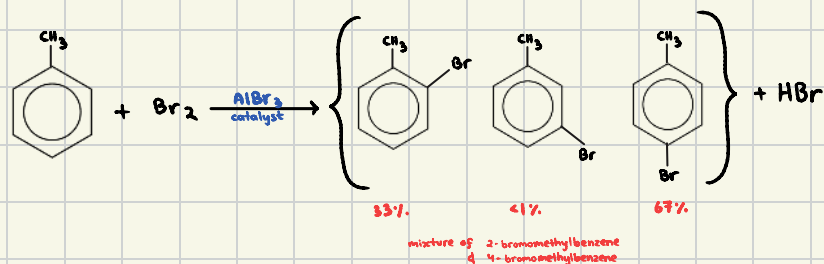


→ bromination of methylbenzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.



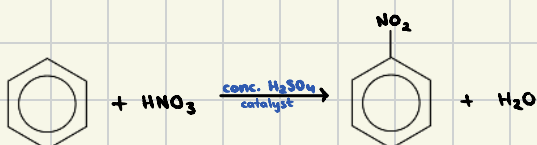
② nitration

→ nitration of benzene

reagent : concentrated HNO_3

catalyst : concentrated H_2SO_4

conditions : temp. 50°C

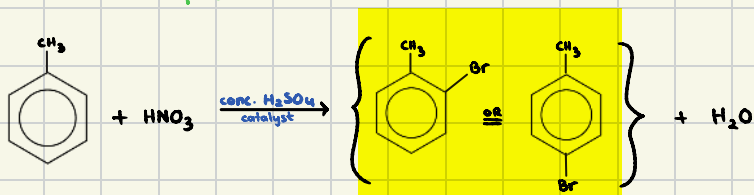


→ nitration of methylbenzene

reagent : concentrated HNO_3

catalyst : concentrated H_2SO_4

conditions : temp. 30°C



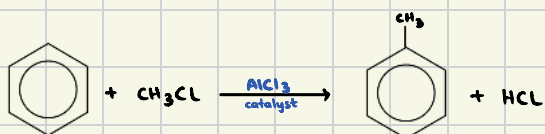
③ Friedel-Crafts Alkylation

→ Friedel-Crafts alkylation of benzene

reagent : CH_3Cl

catalyst : AlCl_3

conditions : anhydrous conditions & heat (50°C)

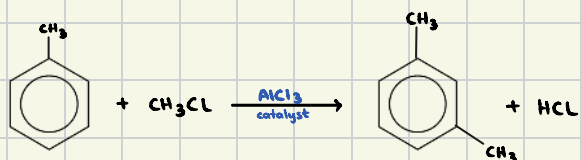


→ Friedel-Crafts alkylation of methylbenzene

reagent : CH_3Cl

catalyst : AlCl_3

conditions : anhydrous conditions & heat (50°C)



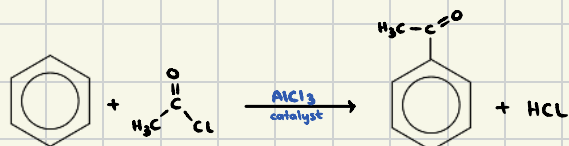
④ Friedel-Crafts Acylation

→ Friedel-Crafts acylation of benzene

reagent : acyl chloride (e.g. CH_3COCl)

catalyst : AlCl_3

conditions : heat (60°C)

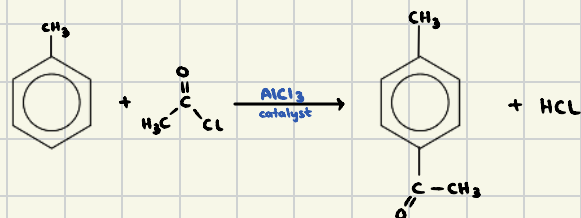


→ Friedel-Crafts acylation of methylbenzene

reagent : acyl chloride (e.g. CH_3COCl)

catalyst : AlCl_3

conditions : heat (60°C)



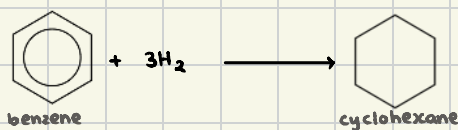
⑤ Hydrogenation

→ hydrogenation of benzene

reagent : H_2

catalyst : Pt/Ni

conditions : heat & pressure



this reaction breaks the delocalised π system of the benzene ring, converting it into a saturated ring

⑥ Complete oxidation of the side chain

sidechain of methylbenzene is OXIDISED by

hot KMnO_4 in alkaline conditions

followed by acidification by dilute acid

